Signal Planning Process

(HANDOUTS)

Signal Officer Advance Course (SOAC) Subject 8.6 • Signal Officer Basic Course (SOBC) Subject 8.5

Courses / Subjects	SOAC – Module 8 (Signal Planning), Subject 8.6 Signal Planning Process SOBC – Module 8 (Signal Plans and Orders), Subject 8.5 Signal Planning Process
References	PAM 6-0102 Signal Operations (15 April 2022): Chapter II, Fundamentals of PA Signal Operations; Chapter III, Planning and Execution PAM 6-00 Command and Control Communications, and Cyber Systems (2nd Edition, 11 December 2017): Chapters II and III PAM 3-00 Army Operations (06 May 2014): Chapter V (mission variables) and Chapter VI (the operations process) Supporting: PAM 3-0201 Commander and Staff Organization and Operations (29 December 2023): Chapters VIII (TLP) and IX (MDMP)
How to use this handout	Read Parts I to IV before the session. Parts V and VI are covered in class. Part VII is the practical exercise scenario, a continuation of Exercise BANTAY LINYA from the MDMP subject. Part VIII holds the worksheets your group completes for the graded planning output and presentation.

Introduction

PAM 6-0102 calls signal plans “the cornerstone of every signal support operation” (para 0217). A signal plan states the signal challenges in a given operational environment and the technical solutions to them, including how communications will be sustained and recovered. The signal planning process is the series of logical steps a signal planner follows so that no part of signal support is missed, and it is carried out **in conjunction with the MDMP, and with troop leading procedures (TLP) for small units** (PAM 6-0102, para 0304). This handout shows how the process fits into the staff’s planning, how to analyze a mission for its communication requirements, how to build a communication concept, and how to put signal support into the unit’s plan or order.

Learning Objectives

Signal Officer Advance Course – Subject 8.6

TLO. At the end of the session, the students will be able to analyze the signal planning process using MDMP/TLP in support of operations. (C3)

ELO	Enabling Learning Objective	Covered in
8.6.1	Analyze MDMP/TLP in signal planning (C4)	Part II
8.6.2	Analyze signal communication requirements based on mission analysis (METT-TC) (C4)	Part III
8.6.3	Develop and analyze the communication concept for operations (C4)	Parts IV, V
8.6.4	Integrate signal support into the operational plan (C4)	Part VI

Signal Officer Basic Course – Subject 8.5

TLO. At the end of the session, the students will be able to perform basic signal planning procedures using TLP/MDMP to support unit communication requirements during operations in accordance with applicable doctrine. (P3)

ELO	Enabling Learning Objective	Covered in
8.5.1	Describe MDMP and TLP in signal planning (C3)	Part II
8.5.2	Identify signal communication requirements based on mission analysis (C2)	Part III
8.5.3	Prepare a basic communication concept for operations (P3)	Parts IV, V
8.5.4	Assist in integrating signal support into the operational plan (P3)	Part VI

Both POIs list the same content: MDMP and TLP in signal planning; mission analysis (METT-TC for communications); the signal planning process and steps; communication concept development; and integration of signal support in operations. SOAC analyzes and develops at the battalion S6 level; SOBC performs the basic procedures as the Signal platoon leader who supports the S6.

Definition of Terms

Term	Meaning
Signal planning process	A series of logical steps that ensures all aspects of signal support operations are systematically addressed; done in conjunction with the MDMP, and with TLP for small units (PAM 6-0102, para 0304).
Signal estimate (C4S estimate)	An orderly and logical examination of all factors, known or assumed, that may affect the accomplishment of the signal support mission; revised as the situation changes (PAM 6-0102, paras 0302–0303).
METAL	Memory aid for the signal estimate: Mission, Enemy signal capabilities, Troops available, Area of operations, Logistical requirements (PAM 6-0102, para 0303).
METT-TC	The mission variables: Mission, Enemy, Terrain and weather, Troops and support available, Time available, Civil considerations (PAM 3-00, para 0524).
C4S	Command and control communications, and cyber systems: the capabilities that enable signal to support Army operations (PAM 6-0102, para 0202).
CEOI / CESI	Communications-electronics operating instructions (“what to do”) and communications-electronics standing instructions (“how to do”) (PAM 6-0102, para 0305e).
PACE	Primary, alternate, contingency and emergency plans – part of the general concept of signal support (PAM 6-0102, para 0312a).
Running estimate	A staff section’s continuous assessment of whether current operations are going as intended and future operations are supportable (PAM 3-00, para 0609).
Annex H (Signal)	The C4S annex to the operation order or plan, prepared by the S6, with its appendices and tabs (PAM 3-0201, para 0229e; PAM 6-0102, para 0314).
Spectrum management	Planning, coordinating and managing the use of the electromagnetic spectrum through operational, engineering and administrative procedures (PAM 6-0102, para 0218).

Part I. Signal Planning in Army Operations

1. Planning within the operations process

All operations follow the operations process: **plan, prepare, execute and assess**, driven by the commander (PAM 3-00, para 0601). A plan is "a design for a future or anticipated operation… a framework from which to adapt rather than a script to follow" (para 0613). Commanders and staffs use the MDMP and TLP to integrate activities during planning (para 0603). The signal plan is the signal part of that design, and it must be ready to adapt when the operation changes.

2. Why signal planning matters

PAM 6-00 states the principle of the **inseparability of operations and C4S**: military operations and military C4S are integral to each other (para 0201b). PAM 6-0102 adds that communication support **should never be the limiting factor** to the commander's ability to execute a mission (para 0209). Signal planning is how the signal officer makes sure it is not.

3. Principles the signal plan must satisfy

Source	Principle	What it asks of the signal plan
PAM 6-0102 (paras 0209–0213)	Operations focused	Support the operation; never let batteries, equipment or lost contact stop the mission
	Interoperable	All units, including other services, can talk to each other
	Redundant	Multiple paths, backups and self-healing strategies
	Scalable	Expands or tailors as the operation moves through its phases
	Secured	COMSEC policy governs every means of communication
PAM 6-00 (para 0201)	Command responsibility	Commanders at all echelons are responsible for their C4S
	Advance provision	Provide C4S ahead of projected operations whenever practicable
	Common procedures, lateral interconnection	Messages can pass from any point in one system to another without major delay
	Reliability, speed and security	Reliability is paramount; weigh speed against security in light of the situation
PAM 6-00 (para 0202)	Survivability, flexibility	Loss of one part does not disrupt the whole; adapts to changing situations

4. Signal operations activities that the plan covers

PAM 6-0102 groups Philippine Army signal support services into four categories: **command and control communication** (fixed, tactical and special purpose communication); **cyberspace operations and electronic warfare**; **signal repair and maintenance**; and **signal recovery and restoration** (para 0131). Under command and control communication, the signal officer's activities are the signal estimate, signal planning, electromagnetic spectrum management, and support to command and control such as operations centers and mobile command centers (paras 0215–0219).

Part II. MDMP and TLP in Signal Planning

The signal planning process does not run on its own. It runs inside the unit's planning: the battalion staff uses the MDMP, while the Signal platoon or section leader uses TLP (PAM 6-0102, para 0304). The table shows what the S6 does at each step of the MDMP, which signal planning step it feeds, and what comes out.

MDMP step (PAM 3-0201)	S6 / signal planner actions	Signal planning step (PAM 6-0102)	Signal output
1. Receipt of mission	• Update the C4S running estimate • Gather tools: higher OPORD and Annex H, CEOI/CESI, SOPs, maps • Check equipment and personnel status	Develop requirements (start)	Updated C4S estimate; signal input to WARNORD 1
2. Mission analysis	• Analyze the mission for communications (METT-TC, METAL) • List specified and implied signal tasks • Determine available resources and shortfalls • Recommend EEFIs; plan reporting for collection assets	Develop requirements; Determine available resources	Signal estimate (facts, assumptions, shortfalls); C4S input to mission analysis brief
3. COA development	• Draft a communication concept for each COA • Place CPs and critical communications nodes on each sketch	Develop priorities; Prepare a tentative plan	Communication concept per COA (PACE)
4. COA analysis (war game)	• Compare C4S needs with available assets for each COA • Test against EW, cyber, terrain and equipment loss	Develop recovery procedures	C4S supportability of each COA; recovery measures
5–6. Comparison and approval	• Rank COAs for C4S supportability • Recommend through the signal estimate	Develop priorities	C4S recommendation; refined priorities
7. Orders production	• Complete detailed plan; site survey and propagation • Prepare Annex H, appendices and tabs; paragraph 5 input	Finalize the plan; Publish plans and orders	Annex H (Signal); CEOI extract; net diagrams

Compiled from PAM 3-0201 Chapter IX and PAM 6-0102 Chapter III

TLP for the Signal platoon or section leader

A Signal platoon leader plans the platoon's part of the battalion signal plan with the eight steps of TLP (PAM 3-0201, para 0812). The same signal planning steps apply on a smaller scale:

TLP step	Signal platoon leader's focus
1. Receive the mission	Read the battalion order and Annex H; identify the platoon's nets, sites and tasks
2. Issue a warning order	Alert operators; start battery charging, equipment checks and loading
3. Make a tentative plan	Requirements, resources, priorities; draft PACE and site plan
4. Initiate movement	Move teams to relay or retrans sites early if needed

TLP step	Signal platoon leader's focus
5. Conduct reconnaissance	Site survey; check line of sight and propagation; signal coverage and gaps
6. Complete the plan	Final site locations, net diagram, recovery procedures
7. Issue the order	Brief operators on nets, call signs, frequencies (CEOI), PACE and COMSEC
8. Supervise and refine	Communication checks and rehearsals; adjust the plan as the situation changes

Part III. Mission Analysis for Communications

1. Sources of signal requirements (PAM 6-0102, para 0305)

- **Commander's guidance or intent** – the concept of operations is a main source; the commander directs the means to be provided and the priorities.
- **Staff guidance** – the other staff officers are the users of signal support, so ask what they need.
- **Higher headquarters, subordinate and adjacent units** – what higher requires of you and what subordinates and adjacent units expect.
- **Standard operating procedures** – a good SOP cuts planning time.
- **CEOI and CESI** – what to employ and how to use it.

2. METT-TC for communications

PAM 3-00 defines the mission variables as METT-TC (para 0524). The signal planner asks how each variable affects communications:

Variable	Questions the signal planner asks
Mission	What are the specified and implied signal tasks? Which nets and links must the unit enter or provide? What does the commander's intent need from C4S?
Enemy	Can the enemy intercept, jam or direction-find our transmissions? Does it attack infrastructure such as cell sites? Does it use drones or cyber means?
Terrain and weather	Where is line of sight blocked? Where are the dead spaces? How do rain, storms and foliage affect propagation and access to sites?
Troops and support available	What radios, generators, batteries, operators and funds do we have? What support can higher or adjacent units give?
Time available	How long to move, install, test and rehearse? Can sites be established before the operation begins?
Civil considerations	Commercial networks, power, the NTC frequency rules, coordination with PNP, LGU and other agencies, crowded events.

3. The signal estimate

The signal estimate examines all factors that may affect the signal support mission. It may be one estimate for all means or separate estimates for each service, and it is revised as the situation changes (PAM 6-0102, para 0302). The memory aid **METAL** helps the planner prepare it:

M	E	T	A	L
Mission in precise terms	Enemy signal capabilities	Troops available	Area of operations	Logistical requirements

The signal estimate is the S6's running estimate. Its format follows the generic running estimate in PAM 3-0201 (Figure 7.1): situation and considerations; mission; courses of action; analysis; comparison; recommendations and conclusions. The signal planner gives the operations officer the items for recommendation or decision in the form of the signal estimate (PAM 6-0102, para 0314).

Part IV. The Signal Planning Process

PAM 6-0102 (paras 0305–0314) describes seven steps:

Step	What the signal planner does	Key output
1. Develop signal and C4S requirements	Obtain and evaluate all information on the signal support and C4S capabilities the unit needs, from the commander, staff, higher/subordinate/adjacent units, SOPs, CEOI and CESI.	List of requirements (who needs what, where, when)
2. Determine available resources	Keep an up-to-date chart of C4S equipment (authorized, on hand, serviceable, unserviceable, reserve), personnel (authorized, actual, training status, shortages, committed) and funds.	Resource status; shortfalls
3. Develop priorities	Compare requirements with capabilities. If resources are not enough, set priorities by signal supportability and importance to the mission.	Prioritized requirements
4. Develop recovery procedures	Plan for interrupted connectivity, electronic warfare, cyberattacks and compromised frequencies.	Recovery and contingency measures
5. Prepare a tentative plan	The most complex step: general concept with PACE; allocation of resources; siting of C4S centers and facilities; security measures; interconnections and alternate routing.	Tentative signal plan (communication concept)
6. Finalize the plan	Check the whole plan: meets requirements, practical, resources adequate, follows priorities, areas of responsibility set, site survey and propagation done, coverage and gaps known, security in place.	Final signal plan
7. Publish the plans and orders	Prepare plans and orders with diagrams and instructions; give the S3 the signal estimate; prepare the C4S annex with its appendices and tabs.	Annex H (Signal); appendices, tabs; paragraph 5 input

PAM 6-0102, paras 0305–0314

Step 2 detail – resource status chart

Resource	Consider
Equipment (para 0307)	Authorized (TE), on hand, serviceable, unserviceable, reserve pool, capabilities of each system
Personnel (para 0308)	Authorized (TO), actual fill-up, training status, critical shortages, committed or deployed, reserve
Funds (para 0309)	Funds for installation, operation and maintenance of C4 systems and administrative needs

Step 4 detail – recovery procedures (para 0311)

Threat to communications	The plan must include
Interrupted connectivity	Redundant communication; equipment repair and replacement; on-site technical support
Electronic warfare	Countermeasures and counter-countermeasures against jamming and interception
Cyberattacks	Monitoring of the PA network, including mobile ad-hoc networks; CSIRT and first-responder procedures
Compromised frequencies	Procedures to change frequencies quickly and efficiently when equipment falls into enemy hands

Step 6 detail – finalize checklist (para 0313)

- ☐ The plan accomplishes the signal support requirements of the unit
- ☐ The plan is practical and resources are adequate
- ☐ The emphasis follows the priorities set in Step 3
- ☐ Areas of responsibility of each C4S center and facility are set
- ☐ A site survey or reconnaissance and propagation check was made
- ☐ Locations of C4S centers and facilities were pinpointed during the survey
- ☐ Signal coverage and gaps were determined
- ☐ Security measures are in place

Part V. Communication Concept Development

The communication concept is the heart of the tentative plan (Step 5). PAM 6-0102 para 0312 lists what it covers. The signal planner develops one for each friendly COA during COA development, tests it in the war game, and completes it for the approved COA.

1. Elements of the communication concept

Element (para 0312)	Questions to answer
a. General concept with PACE	For each critical link, what are the primary, alternate, contingency and emergency means? When do users switch?
b. Allocation of resources	Which unit gets which radios, operators, generators and batteries?
c. Siting of C4S centers and facilities	Where do CPs, relay or retrans sites go, based on requirement density, unit dispositions, support units, terrain, propagation, projected operations, accessibility and logistics?
d. Security measures	COMSEC, EEFIs, site security, emission control
e. Interconnections and alternate routing	How do the nets connect to higher, adjacent and supporting units? What is the alternate route if a link or site is lost?

2. Building a PACE plan

Build PACE for each critical communication requirement, not for the unit as a whole. Each means should be different and independent of the others, so the loss of one does not take down the next.

	Means	Example for a battalion command net
P – Primary	Normal means for routine traffic	VHF FM combat net radio
A – Alternate	Different means, ready to take over	HF manpack
C – Contingency	Slower or less capable, but works	Special purpose communication (commercial cellular or satellite phone), under COMSEC policy
E – Emergency	Last resort	Messenger, liaison officer, pre-arranged visual signals

3. Analyzing the concept

SOAC students analyze each concept, and SOBC students check their basic concept, against these tests:

- **Requirements:** does every requirement from Step 1 have a means, in priority order?
- **Principles:** operations focused, interoperable, redundant, scalable, secured (PAM 6-0102); reliable, survivable, flexible (PAM 6-00).
- **Terrain and weather:** are coverage gaps covered by relay, retrans, HF or another means?
- **Threat:** does the concept survive interception, jamming, cyberattack, site attack and compromised frequencies?
- **Resources:** do the radios, operators, power and funds add up? What is the shortfall and who fills it?
- **Phasing:** does the concept scale as the operation moves through its phases?

4. Special situations

Joint operations: use procedures such as the vigilance net, lending equipment to another service, or embedding radio operators (PAM 6-0102, paras 0315–0319). **Coalition operations:** identify the

points of integration between national networks (paras 0320–0322). **HADR:** keep HADR nets separate from the Army's secured operational net; deploy signal teams and mobile command centers (paras 0323–0326).

Part VI. Integration of Signal Support in Operations

Signal support is integrated into the operation through the staff, not added after. The signal planner does this in four ways:

- **Through the signal estimate:** the signal planner recommends to the operations officer what to include in the unit's plan or order, in the form of a signal estimate (PAM 6-0102, para 0314).
- **Through the COA sketches and war game:** CPs and critical communications nodes appear on every COA sketch, and the S6 assesses C4S feasibility of each COA (PAM 3-0201, paras 09116, 09155).
- **Through the order:** the S6 prepares Annex H (Signal) with its appendices and tabs, and contributes to Annex C and Appendix 12 (Cyberspace Electromagnetic Activities) with the S3 and S2 (PAM 3-0201, para 0229e).
- **Through preparation and execution:** communication checks, rehearsals and supervision (TLP Step 8), and the C4S running estimate kept current through the operation (PAM 3-00, para 0609).

Paragraph 5 (Command and Signal) – signal content

The base order's paragraph 5 summarizes the signal plan for the whole unit. At a minimum, state:

- Location of the command post and alternate command post
- Primary and alternate means to higher headquarters, and PACE for critical nets
- CEOI index and edition in effect
- Restrictions such as emission control or listening silence
- EEFIs and COMSEC instructions
- Reporting of loss of communications

Suggested Annex H outline for the practical exercise

Use your unit SOP format where one exists. For the exercise, groups use this outline:

Paragraph	Content
1. Situation	Enemy signal capabilities; friendly signal support from higher and adjacent units; attachments; assumptions
2. Mission	Signal support mission of the unit
3. Execution	Communication concept (PACE); tasks to subordinate units and the Signal platoon; site locations; coordinating instructions
4. Sustainment	Batteries, generators, maintenance and repair, replacement of equipment
5. Command and Signal	Signal staff and CP locations; recovery procedures; reporting
Appendices / tabs	Net diagram; PACE matrix; CEOI extract; site plan

Self-Check Questions

1. What is the signal planning process, and with which planning procedures is it conducted?
2. List the seven steps of the signal planning process in PAM 6-0102.
3. What are the five sources of signal requirements?
4. What does the memory aid METAL stand for, and what is it used for?
5. Give one communication question for each of the METT-TC variables.
6. Which three resource areas does the signal planner track in Step 2?
7. Name the four contingencies that recovery procedures must cover.
8. What five elements make up a tentative signal plan?
9. Explain PACE and why each means should be independent of the others.
10. Name the five principles of signal operations in PAM 6-0102.
11. In which MDMP step does the S6 compare C4S needs with available assets for each COA?
12. What does the signal planner give the operations officer in Step 7, and what annex does the S6 prepare?

Summary

Signal planning is how the signal officer keeps communication support from becoming the limiting factor in an operation. The signal planner follows seven steps – develop requirements, determine resources, develop priorities, develop recovery procedures, prepare a tentative plan, finalize the plan, publish the plans and orders – inside the unit's MDMP or TLP. Mission analysis for communications applies METT-TC and the METAL signal estimate. The communication concept sets PACE for each critical link, sites C4S facilities, secures the network and plans alternate routing. The plan is integrated into the operation through the signal estimate, the COA sketches and war game, Annex H and paragraph 5, and supervision during preparation and execution.

References

- PAM 6-0102 Signal Operations, promulgated 15 April 2022
- PAM 6-00 Command and Control Communications, and Cyber Systems, 2nd Edition, 11 December 2017
- PAM 3-00 Army Operations, 06 May 2014
- PAM 3-0201 Commander and Staff Organization and Operations, 29 December 2023 (supporting)
- Program of Instruction, Signal Officer Advance Course, Subject 8.6, and Signal Officer Basic Course, Subject 8.5